Introduction

The F distribution arises as the ratio of two independent scaled chi-squared random variables. It is the reference distribution for the F-statistic in ANOVA, for tests of variance equality between two normal samples, and for omnibus tests in linear regression. Its shape is controlled by two degree-of-freedom parameters, one for the numerator and one for the denominator.

Prerequisites

Chi-squared distribution.

Theory

If $U \sim \chi^2_{d_1}$ and $V \sim \chi^2_{d_2}$ are independent, then

$$F = \frac{U/d_1}{V/d_2} \sim F_{d_1,d_2}.$$

The numerator df is d_1; the denominator df is d_2.

Moments (when they exist):

- $E[F] = d_2/(d_2 - 2)$ for $d_2 > 2$.
- $\mathrm{Var}(F) = \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$ for $d_2 > 4$.

Relationships to other distributions:

- $1/F_{d_1,d_2} \sim F_{d_2,d_1}$.
- $t_\nu^2 \sim F_{1,\nu}$.
- As $d_2 \to \infty$, $d_1 \cdot F_{d_1,d_2} \to \chi^2_{d_1}$.

One-way ANOVA: with k groups and n total observations, the F-statistic

$$F = \frac{\mathrm{MS}_{\mathrm{between}}}{\mathrm{MS}_{\mathrm{within}}}$$

follows $F_{k-1,n-k}$ under the null of equal means.

Variance ratio test: for two independent normal samples with variances $\sigma_1^2 = \sigma_2^2$,

$$F = s_1^2/s_2^2 \sim F_{n_1-1,n_2-1}.$$

Assumptions

Independence between U and V; normality of the underlying data for the derived sample statistics.

R Implementation

```r
# PDF visualisation
x <- seq(0, 5, length.out = 400)
plot(x, df(x, 5, 20), type = "l", col = "#2A9D8F", lwd = 2,
     main = "F(5, 20) density", ylab = "f(x)")
lines(x, df(x, 5, 5),   col = "#F4A261", lwd = 2)
lines(x, df(x, 20, 20), col = "#6A4C93", lwd = 2)
legend("topright", c("F(5,20)", "F(5,5)", "F(20,20)"),
       col = c("#2A9D8F", "#F4A261", "#6A4C93"), lwd = 2)
```

```r
# Verify construction
n <- 1e5; d1 <- 4; d2 <- 10
U <- rchisq(n, df = d1); V <- rchisq(n, df = d2)
F_sim <- (U/d1) / (V/d2)
c(emp_mean = mean(F_sim), theoretical = d2 / (d2 - 2))

# F-statistic from ANOVA
set.seed(2026)
df <- data.frame(y = c(rnorm(30, 50, 8), rnorm(30, 55, 8), rnorm(30, 53, 8)),
                 g = factor(rep(c("A", "B", "C"), each = 30)))
anova(lm(y ~ g, data = df))
```

Output & Results

```
emp_mean theoretical
    1.252       1.250

Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value  Pr(>F)
g          2  394.7   197.3    3.57 0.03218 *
Residuals 87 4811.2    55.3
```

The empirical F ratio recovers the theoretical mean, and the ANOVA F-statistic of 3.57 on (2, 87) df has p-value 0.032.

Interpretation

Almost every ANOVA and linear-model-comparison result in applied statistics is reported as an F-statistic with numerator and denominator df. Reporting "F(2, 87) = 3.57, p = 0.032" is a standard form.

Practical Tips

- Order the df correctly: F(df1, df2) is not the same as F(df2, df1) in general.
- For two-sided variance tests, some software uses max/min of s_1^2/s_2^2 to avoid ambiguity.
- Non-central F (with non-centrality parameter λ) governs power calculations for ANOVA.
- The F-distribution is highly sensitive to non-normality; for skewed data prefer Welch's ANOVA or non-parametric alternatives.
- `qf(0.95, d1, d2)` gives the critical value for a one-sided 5% test.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem

- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots